SLJSO 2023 Solutions

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48	Algebra	
49	Algebra	
50	Algebra	

Question 1

This question requires knowledge of the composition of biological molecules. Let's focus on a couple points to help us decipher the other options:

- ALL amino acids contain nitrogen; they have a central carbon bonded to an amino group (NH_2)—where the nitrogen comes from, a carboxyl group (COOH), and a side chain (R group that is different for every amino acid).
- The molecular formula for cellulose is $(C_6H_{12}O_5)_n$ the only elements included are carbon, hydrogen, and oxygen. There is no nitrogen in cellulose. The molecular formula for starch is $(C_6H_{10}O_5)_n$; therefore, starch also does not contain nitrogen.
- Membrane proteins contain nitrogen because, as proteins, they are made of amino acids. Amino acids, as we've already established, contain nitrogen.
- Enzymes are proteins, so they contain nitrogen as well.
- It is important to remember that the question says ALWAYS contains nitrogen. Some very specific lipids contain nitrogen, but the general population of lipids contains hydrogen, carbon, and oxygen. Therefore, lipids are not correct for this.
- Anything DNA, mRNA, or tRNA related will have nitrogen because all nucleotides (adenine, cytosine, guanine, thymine, and uracil) contain a nitrogenous base.

Now, let's see which option is correct:

A) Incorrect because cellulose does not contain nitrogen.

B) Incorrect because most lipids do not contain nitrogen.

C) CORRECT because enzymes, mRNA, and tRNA all contain nitrogen

D) Incorrect because starch does not contain nitrogen.

Correct Answer: C

Question 2

This question requires a basic understanding of denatured proteins (enzymes are proteins). Proteins in too high of a temperature will lose their original secondary, tertiary, and quaternary structures due to the hydrogen bonds and disulfide bridges between their components weakening and breaking. The rule of thumb is if the temperature is beyond the natural temperature range of a bacteria, they will die out. Also, going a little over the temperature range will not mean that the entire population dies in an instant; it happens gradually. Let's see which diagram fits these trends, remembering that enzyme X ranges from 5 to 20 degrees and enzyme Y ranges from 40 to 85 degrees:

A) This graph is inaccurate because the population declines too rapidly. Also, enzyme Y does not start from 0 degrees, which is a red flag because both enzymes are put in at the same time. Enzyme Y starts slower, but it is not non-existent.
B) This graph is accurate because both enzymes start at 0, and X tapers off at a reasonable spot given the temperature range. Y also tapers off at a reasonable spot (past 85 degrees).
C) Inaccurate because the X enzyme should taper off; it cannot survive that much past 20 degrees.
D) Inaccurate because both populations do not taper off where they should. This is not a linear process.

Correct Answer: B

Question 3

Facilitated diffusion is the process by which molecules cross specific membrane proteins or channels without ATP input (not active). Diffusion is also key to solving this problem, because it is based on concentration gradients. When the inside of the cell is relatively similar to that of the outside the cell, there will be no net movement.

a) Inaccurate because there can be more X outside the cell, but if it is equal to the X inside the cell, there is no need for more transport.
b) Yes, there is a certain amount of carrier proteins on the cell membrane, but it usually is not the cause of plateauing transport
c) Accurate because as soon as inside the cell = outside of the cell in terms of concentration, net movement is 0, and there is no need to disrupt this balance

Correct Answer: D

Question 4

This question requires knowledge of mitosis (and why it differs from meiosis). Mitosis is purely for asexual purposes, not for creating gametes. Mitosis is used for growth and repair, maintenance, and reproduction in some asexual organisms. The purpose is to create two genetically IDENTICAL daughter cells (clones). The KEY to this question is the phrase "single-celled organisms." Single-celled organisms are not trying to become multicellular organisms by producing genetically identical cells or exhibiting growth; their sole purpose of mitosis is to reproduce and create more identical single-celled organisms.

Correct Answer: D

Question 5

In this case, tissue fluid (interstitial fluid) is what is being given to body cells as a medium for nutrient and gas exchange. Therefore, phagocytes (for immune purposes), proteins, and sodium will ALL be present in tissue fluid.

Correct Answer: A

Question 6

To solve this problem, don't forget the concept of polarity. Positively charged compounds will bond with negatively charged compounds as opposed to other positively charged compounds (in vice versa). The carboxyl group (COOH) will give up a proton to form COO^- and the amino group (NH_3) will gain a proton to form NH_4^+.Therefore, we can easily eliminate option D because two negatively charged groups will not attract each other. The rest of the options are okay when it comes to fulfilling polarity. Peptide bonds typically form between two amino acids in a lateral (as opposed to vertical) form. Therefore, option A best forms a lateral bond between these two amino acids.

Correct Answer: A

Question 7

Red blood cells (erythrocytes) have the primary function of carrying oxygen due to their hemoglobin contents. Keeping that in mind, the right atrium contains deoxygenated blood (about to go to the right ventricle to pick up oxygen from the lungs), and the left atrium contains oxygenated blood (came back from the lungs). If there is a hole between the left and right atria, the blood will get mixed between them. Mixing deoxygenated with oxygenated blood will decrease overall oxygen levels. Having lower oxygen levels calls for higher hemoglobin levels (and thus red blood cells) to capture more oxygen.

Correct Answer: D

Question 8

If you are not familiar with water potential, we will rephrase this question to talk about osmolarity. Areas with higher (less negative) water potential simply have more water and less solute. Areas with lower (more negative) water potential simply have less water and more solute. Remember that water always wants to go from areas with lower to higher solute concentration. For this scenario, the solution the cell is in has a higher water potential (more water and fewer solutes) than the inside of the cell. Therefore, water will want to enter the cell, which will cause the central vacuole to expand and the cell to become more turgid (swollen). Option C fits the best because it is clearly swollen due to the cell wall getting rounder, BUT not D because the water potential is only slightly higher. Option A does not fit because the vacuole did not expand, and option B does not fit because plasmolysis is what would happen if the osmolarity was flipped.

Correct Answer: C

Question 9

By "reducing water potential gradient," these plants try to adapt to prevent water loss. Let's look at each option to see if they help with water conservation:

a. Accurate, because rolling up means only one side of the leaf will be exposed to sunlight, which reduces water loss.
b. Accurate, having hairy leaves (trichomes) reduces water loss by trapping water and preventing evaporation.
c. Accurate, sunken stomata reduce water loss by creating a humid pocket of air to reduce evaporation.
d. Inaccurate, yes, having fewer gaps to lose water means less evaporation, BUT it does not change the water potential gradient (difference between water concentration inside and outside the plant).
e. Inaccurate, yes, having fleshy leaves is an indicator of a xerophytic plant storing water, BUT it does not change the water potential gradient.

Correct Answer: B

Question 10

A) Accurate, the diameter of the blood vessels directly influences their blood pressure. As diameter increases, blood pressure also increases.
B) The number of RBC in circulation only affects the oxygen levels of the blood. That shift is too microscopic to directly affect blood pressure.
C) The systolic pressure of the heart ventricles is simply the amount of force they can generate when ejecting blood. That force is directly proportional to blood pressure (higher force = higher pressure).
D) The volume of blood directly affects blood pressure. If more blood is trying to fit in the same vessel, that vessel will face more pressure.

Correct Answer: B

Question 11

A) The waxy cuticle has nothing to do with water uptake from the soil. It has to do with water retention and preventing water loss, which is different from uptake from the soil.

B) Large numbers of mitochondria have nothing to do with water uptake from the soil. Water uptake from the soil is done by root hair cells using osmosis.

C) This is referring to the root hair cells, which indeed have everything to do with water uptake from the solution. Their elongated structure increases the surface area to volume ratio to increase absorption rates.

D) Though cell walls are permeable to water, they are not the primary mechanism for water uptake from the soil.

Correct Answer: C

Question 12

If this sounds like a lot of information, it can be very helpful to draw a flow chart or mind map to demonstrate the nature of these interactions in chronological order. In the usual situation (with function HDP), this is our flowchart:

Plasmodium → Infects RBC → Breaks Hemoglobin into → Free heme groups (toxic to plasmodium) → Plasmodium converts them to non-toxic crystalline hemozoin using HDP (enzyme)

With malaria drug (inhibited HDP):

Plasmodium → Infects RBC → Breaks Hemoglobin into → Free heme groups (toxic to plasmodium) → Plasmodium is now unable to convert to non-toxic hemozoin

Hopefully that clears up any confusion. Let's consider the options:

A) Inaccurate, there will be an increase in free heme groups because they are not being converted to hemozoin.
B) Inaccurate, inhibiting HDP does not directly affect hemoglobin production, just toxic heme conversion to non-toxic hemozoin.
C) Inaccurate; if free heme groups are toxic to Plasmodium and they are no longer being converted to their non-toxic form, Plasmodium will find it harder to survive and reproduce, not easier.
D) Accurate, plasmodium will find it harder to reproduce due to the buildup of toxic heme groups

Correct Answer: D

Question 13

The simplest solution to this problem is to look for the letter that all the other letters end up at (regardless of the path they take). This is because decomposers are at the end of every food chain, irrespective of how long that food chain is. Whether it was a primary consumer that died or an apex predator that died, whether it was an herbivore or a carnivore that died, they all end up decomposing. That letter is D because producers grow from the work done by decomposers; letter A can die and get decomposed right away, or it can be consumed by a secondary (B) or tertiary organism (C), which will eventually die and get decomposed as well.

Correct Answer: D

Question 14

Let’s define each term:

a. Habitat is the environment an organism lives in (including both biotic and abiotic factors).
b. A niche is the role that an organism plays, including what it eats, where it lives, and other environmental actions (it encompasses everything the organism does).
c. Trophic level (focuses on who eats who specifically)

This scenario is only focusing on trophic levels, not habitat or niche.

Correct Answer: D

Question 15

The correct answer here is something that can thrive without oxygen (because this is in a waterlogged garden). The correct answer is option B, denitrifying bacteria (converts beneficial nitrates in the soil to nitrogen gas), because they are anaerobic organisms.

Correct Answer: B

Question 16

The hardness of water is determined by the amount of calcium and magnesium dissolved in the water. Therefore, for one of these compounds to not cause permanent hardness in water, the salt must be insoluble, because otherwise the amount of calcium or magnesium dissolved in the water will increase as a result of adding the compound. As compounds A, B and D are all soluble in water, compound C is the only one that does not cause permanent hardness, so the answer is C.

Correct Answer: C

Question 17

For a compound to have both ionic and covalent bonds, the compound would have to be a salt consisting of at least one polyatomic ion. Compounds A and B are molecular compounds and therefore lack ionic bonds. Compound D is a salt but has only monoatomic ions and therefore lacks covalent bonds. Compound C is a salt with the polyatomic cation NH_4^+, so it will have both ionic and covalent bonds. The correct solution, therefore, is C.

Correct Answer: C

Question 18

For this question, the symmetry of the molecule has to be considered when answering the question. Moving the double bond 1 or 2 carbon atoms to the right will indeed give us a new compound, but moving the double bond 3 carbon atoms to the right will give us the same compound as moving it 1 space would. Moving it 4 spaces gives us the given structure again. This means that B is the correct answer, as only 2 distinct new compounds can be formed by moving the double bond.

Correct Answer: B

Question 19

A useful trick to answer this question is to simply draw out every possible option (should only be a maximum of 4 drawings). By keeping the $CH_2 = CH$ group at the left end of the molecule the same in every drawing, you prevent making any stupid mistakes. You can first start by taking the rightmost carbon from the given structure and attaching it as a methyl group. This can be done on either the second or third carbon atom from the right in the given pentene structure, giving us the first two structures. Furthermore, we can take the two rightmost carbon groups and attach both as methyl groups in a butene structure by attaching them both to the second carbon from the right in this structure. If you try to attach these two carbons somewhere as an ethyl group, you will quickly find that this makes the same compound that you made by attaching one methyl group to the middle atom in the pentene structure. Drawing out all the options gives you the correct answer to this question, which is that there are 3 different structures possible.

Correct Answer: C

Question 20

This is a very simple bit of calculational work. We will start by determining the amount of moles of $CuSO_4 \cdot 5H_2O$ in 0.50g: Molar mass $= 63 + 32 + 4 \times 16 + 10 \times 1 + 5 \times 16 = 249\text{g mol}^{-1} \rightarrow \text{n} = \frac{0.50\text{g}}{249\text{g mol}^{-1}} = 2.0 \times 10^{-3}\text{moles}$ The amount of moles of water that evaporate will be 5 times this amount, which allows us the calculate the mass: $\text{n}_{\text{water}} = 5 \times 2.0 \times 10^{-3}\text{moles} = 1.0 \times 10^{-2}\text{ moles} \rightarrow \text{m}_{\text{water}} = 1.0 \times 10^{-2}\text{ moles} \times 18\text{g mol}^{-1} = 0.18\text{g}$. As you can see, this gives C as the correct answer.

Correct Answer: C

Question 21

The chemical formula of glucose is $C_6H_{12}O_6$. Combining two molecules of glucose while splitting of a molecule of water gives the following chemical reaction: $2C_6H_{12}O_6 \rightarrow H_2O + C_{12}H_{22}O_{11}$. As you can see from this reaction, the correct answer to this question is option B.

Correct Answer: B

Question 22

This question might seem a bit daunting at first, especially if you don't have the periodic table memorized and might not know the atomic number of phosphorus. However, you don't need any knowledge of the atomic number to solve this question. All you need is a simple rule known as Hund's rule. If there are multiple atomic or molecular orbitals with the same energy in a given system (an atom or molecule), the electrons will be divided in such a way that every orbital with equal energy gets 1 of 2 electrons first, followed by filling up the orbitals with any remaining electrons. As you can see, answers A, B, and C do not abide by this rule, as they have the 3rd 3p orbital empty while one of the 3p orbitals is completely filled. The only answer that follows Hund's rule is answer D, so this must be the correct option.

Correct Answer: D

Question 23

I hope that you will be able to figure this question out without my guidance, but here comes the answer, nonetheless. This question is just a fill-in exercise. By reading the question, you will realize that the atomic mass can be found by the following calculation:

$24 \times 0.60 + 25 \times 0.25 + 26 \times 0.15 = 24.55 \text{ amu}$, which gives option D as the correct answer.

Correct Answer: D

Question 24

To solve this question, we must determine the ratio of carbon to oxygen in both compounds. For compound X, the easiest way to obtain this ratio is by multiplying the mass of both elements by two. This gives a ratio of C:O of 12 g:32 g, which, accounting for the mass of the atoms, gives a 1:2 ratio. For compound Y we obtain the ratio by multiplying the mass of both atoms by 4; this gives C:O of 36 g:96 g, which, accounting for atomic mass, gives a ratio of 3:6, which is the same as 1:2. The ratio of carbon atoms to oxygen atoms is the same in both compounds, and given that the molecular weight of Y is higher than that of X, it is possible that Y is a dimer of X, giving C as the correct answer.

Correct Answer: C

Question 25

To solve this question, you start by determining the pressure because of the oxygen gas in the chamber. This is equal to the total pressure minus the pressure of water vapor, which is given:

$p = 745.8\text{mmHg} - 23.8\text{mmHg} = 722\text{mmHg} = \frac{722\text{mmHg}}{760} = 0.95 \text{ atm}$

Filling this into the given formula and solving for n gives:

$n = \frac{pV}{RT} = \frac{0.95 \text{ atm} \times 0.10\text{l}}{0.0821\text{l} \times \text{atm}/(\text{K} \times \text{mol}) \times 298.15\text{K}} = 3.9 \times 10^{-3}\text{moles.}$

Therefore, the solution is option C.

Correct Answer: C

Question 26

To solve this question, we calculate the molecular weight of glucose to determine the number of moles and multiply that by 6 times Avogadro's constant to find the number of oxygen atoms (as that will be equal to 6 times the number of molecules of glucose). This gives:

Molar weight $= 6 \times 12 + 12 \times 1 + 6 \times 16 = 180\text{g mol}^{-1} \rightarrow n = \frac{9\text{g}}{180\text{g mol}^{-1}} 0.05 \text{ moles} \rightarrow \text{number of atoms} = 0.05 \text{ moles} \times 6{,}02 \times 10^{23} \times 6 = 1.8 \times 10^{23}$. The correct answer should therefore be the option A.

Correct Answer: A

Question 27

Looking at the different data points that we are given, it becomes clear that the color intensity increases linearly with concentration and that the color intensity is equal to 10 times the concentration of the solution. A color intensity of 0.025 will give a concentration of $\frac{0.025}{10} = 0.0025\text{M}$, so the correct answer is B.

Correct Answer: - B

Question 28

Note: The atomic weight of calcium is given incorrectly in this question. The weight should be equal to 40 amu.

Another bit of relatively simple calculational work. We will start by determining the molar mass of CaC_2, to determine the amount of moles used. Knowing that CaC_2 reacts 1:1 into acetylene, we can determine the mass of acetylene that should have formed to calculate the yield. This gives:

$$\text{Molar mass} = 40 + 2 \times 12 = 64\text{g mol}^{-1} \rightarrow \text{n} = \frac{32 \times 1000\text{g}}{64\text{g mol}^{-1}} = 500 \text{ moles}$$

$$\text{Molar mass of acetylene} = 2 \times 12 + 2 \times 1 = 26\text{g mol}^{-1} \rightarrow \text{m}_{\text{theoretical}} = 500 \text{ moles} \times 26\text{g mol}^{-1} = 13000\text{g} = 13\text{kg}$$

$\text{Yield} = \frac{11.5\text{kg}}{13\text{kg}} \times 100\% = 88\%$, so option D is correct.

Correct Answer: D

Question 29

To solve this problem, we will first calculate the number of moles of HCl in a 500mL 5M solution and then use this to determine the amount of concentrated HCl needed to reach this amount.

$n_{HCl} = 5\text{M} \times 0{,}500\text{L} = 2{,}5 \text{ moles} \rightarrow V_{required} = \frac{2{,}5 \text{ moles}}{11\text{M}} = 0{,}227\text{L} = 227\text{mL}.$

This gives answer C as the right answer to this problem.

Correct Answer: C

Question 30

The final chemistry problem is luckily quite a simple one. A homogeneous mixture of compounds is one in which all compounds are in the same state (gas, liquid, or solid), while in a heterogeneous mixture they are in different states. As air is a mixture of gases at room temperature, statement B is the correct statement and therefore the correct answer. It is obvious that A and B cannot be true at the same time, so A cannot be correct. Answer C is incorrect because the different gases that make up air have different boiling points, so if you cool down air, there will be a point where some of its components condense into a liquid while others stay a gas. Finally, answer D is incorrect because heating air up will not change the fact that the compounds stay a gas, so air is just as homogeneous of a mixture at 85 degrees Celsius as it is at room temperature.

Correct Answer: B

Question 31

A lot of students confuse this type of question by not converting the units. So, first we must convert the speed to the base units.

$72\ \text{kmh}^{-1} = \frac{72}{3.6}\text{ms}^{-1} = 20\text{ms}^{-1}$

Since we know the initial velocity, the final velocity and the deceleration, we could apply the equation $v^2 = u^2 + 2as$.

Substituting v = 0, u = 20, a = -2, we get s = 100m.

Correct Answer: C

Question 32

According to the conservation of momentum, the final momentum should be equal to the initial momentum. As the system was initially at rest, the initial momentum was 0. So, the final momentum should be 0 as well.

$\text{Final Momentum} = m_P v_P + m_Q v_Q$

Let us assume that velocity is positive to the right direction and negative to the left direction.

Substituting $m_P = m, v_P = -1\text{ms}^{-1}$ and $m_Q = 2m$, we get:

$0 = -m + 2m * v_P$

Hence, $v_P = +0.5\text{ms}^{-1}$

The positive sign of the answer indicates that the object is moving towards the right.

Correct Answer: D

Question 33

When two resistors are connected in parallel, their equivalent resistance can be denoted by the equation $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$.

In this case, $\frac{1}{R_{eq}} = \frac{1}{6\Omega} + \frac{1}{3\Omega} = \frac{1}{2\Omega} \Rightarrow R_{eq} = 2\Omega$

When two resistors are connected in series, their equivalent resistance be denoted by the equation $R_{eq} = R_1 + R_2$. In this case, the two 6Ω and 3Ω resistors (connected in parallel to each other) act as a 2Ω resistor connected in series to the 2Ω resistor on the right. $R_{eq} = 2\Omega + 2\Omega = 4\Omega$

Now we can apply the well-known equation $V = IR$. Rearranging this equation, we get $I = \frac{V}{R}$. Substituting $V = 4V$ and $R = 4\Omega$, we get $I = \frac{V}{R} = \frac{4V}{4\Omega} = 1A$.

Correct Answer: D

Question 34

Talking about the resultant force, the two 50N forces cancel out as they are in opposite directions. However, the two 40N forces acting on the object are in the same direction. So, these two forces add up to generate a force of 80N on the object. Talking about the resultant moment (also known as torque), the two 40N forces exert an equal torque but in opposite directions. This is because the 40N force acting on the top exerts a clockwise moment and the 40N force acting on the bottom exerts a counterclockwise moment, which means they cancel out. However, the two 50N forces both exert counterclockwise moments, which results in the two moments adding up to give a net moment for the object.So, in this situation the object has both a linear acceleration and an angular acceleration.

Correct Answer: D

Question 35

According to the conservation of energy, the final energy must be equal to the initial energy + energy wasted to the surroundings.

Energy at Point P = Energy at point Q + Work done against friction.

Kinetic energy at P + Potential Energy at P = Kinetic Energy at Q + Potential Energy at Q + Work done against friction.

Substituting values, we get;

Kinetic Energy at P + (Potential Energy at P - Potential Energy at Q) = 70 kJ + 15kJ

Kinetic Energy at P + 40kJ = 85kJ

Kinetic Energy at P = 45kJ

Correct Answer: D

Question 36

In questions like this, you must be able to understand what truly happens. If you look very carefully, you will realise that the temperature is constant from t = 4min to t = 7min. This happens when the physical state of the material is changing. In this case, since the material was initially a solid, the change must be the solid melting to become a liquid.

Now, we will look at each individual response and see which one is correct.

(A) - False. The solid starts melting after 4 minutes and completely melts after 7 minutes.

(B) - False. The specific heat capacity of a substance is defined to be the amount of energy needed to raise the temperature of 1kg of a block by 1K. So, when the specific heat capacity is high, more energy is required to heat the substance. In this case, in the solid state, the material heats at a slower rate than in the liquid state. Since the energy supplied to the system is constant, the specific heat capacity of the solid is higher than that of the liquid.

(C) - True. All the material (1kg) completely melts in 3 minutes. In 3 minutes, 3000*3 = 9000J is supplied to the material. So, the latent heat of the material is 9000J.

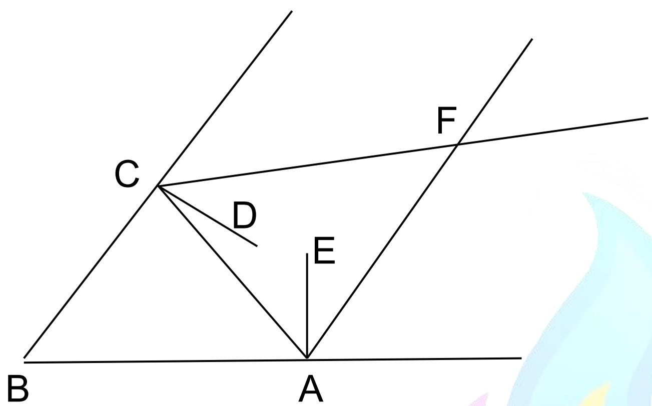
(D) - False. After 7 minutes, the solid has completely melted. However, if the material has evaporated, then there must be another phase where the temperature is constant. But since there is no such phase shown in the graph, we can conclude that the material hasn't evaporated by 10 minutes.

Correct Answer: C

Question 37

This question resembles a simple geometry problem that includes the law of reflection.

The law of reflection states that the angle of incidence equals the angle of reflection.



$\angle ABC = 60°$. Assume $\angle FAE = x$. Then, by the law of reflection, $\angle EAC$ will also be equal to x. Since, $\angle EAB$ equals 90°, $\angle CAB = \angle EAB - \angle EAC = 90-x$.

Since, CBA is a triangle and the sum of angles of a triangle is 180°, $\angle BCA$ will be equal to $180- \angle CBA - \angle CAB = (90-x + 60) = 30+x$.

So, $\angle ACD$ will be equal to $\angle BCD - \angle BCA = 90 - (30+x) = 60-x$.

By the law of reflection, $\angle DCF$ will also be equal to $\angle DCA = 60-x$.

If we consider the triangle ACF, $\angle CAF = 2x$ and $\angle ACF = 2*(60-x) = 120 - 2x$.

So, the angle $\angle AFC = 180 - \angle ACF - \angle CAF = 180 - (2x + 120 -2x) = 180-120 = 60°$.

The angle we are supposed to find is equal to $\angle AFC$ due to the laws of geometry. So, the answer is 60 degrees.

Correct Answer: D

Question 38

For Car A;

$v = u + at$

Substituting $v = 10ms^{-1}, u = 0, t = 5s$, we get $a = 2ms^{-2}$.

$s = ut + \frac{1}{2}at^2$

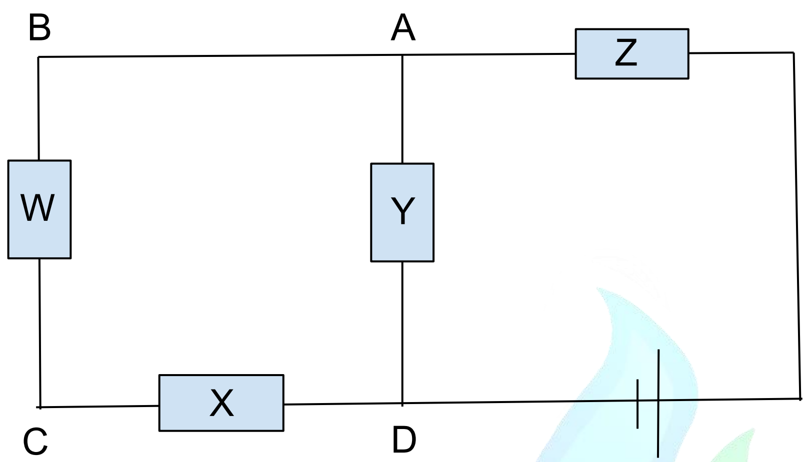
Substituting $u = 10ms^{-1}, t = 2s, a = -2ms^{-2}$, we get s = 16m.

Car B is travelling with a uniform velocity, which means that the acceleration is zero. So, we can apply the equation $s = ut$ to find the distance travelled. Substituting $u = 6m$ and $t = 2s$, we get 12m. The difference in this distance is 16m - 12m = 4m.

Correct Answer: A

Question 39

In this circuit, we can think of light bulbs as resistors with equal resistance, R. Then, we can draw the circuit as follows:



If the current going through the Resistor Z is I, that current divides into two different currents in the junction A. The voltage drop in the path through the resistor Y and through the resistors W and X is the same. The resistance in the path through the resistors W and X is twice the resistance in the path through the Y resistor. So, according to the V=IR law, the current through the resistors W and X will be half of the current through the resistor Y.

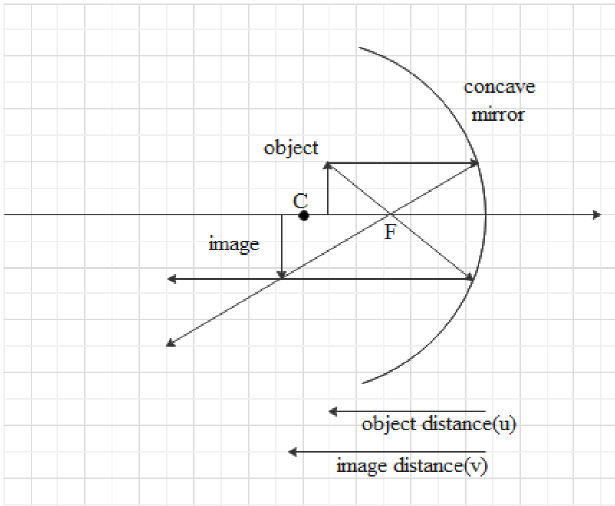
Power $= I^2 R$. Since R is the same for all resistors, Power in resistors W and X < Power in Y < Power in Z.

- Statement I is true since they both have the same power.
- Statement II is false since Z has a higher power than Y.
- Statement III is true since Y has a higher power than X.

Correct Answer: D

Question 40

A concave mirror is a curved mirror with its reflecting surface on the inner side of the curve. It is also called a converging mirror because it focuses parallel light rays to a single point.



As you can see, when the object is between C and F, the image is inverted, magnified, and real. You can draw this ray diagram like so:

1. Draw a straight line parallel to the principal axis. When that light ray hits the concave mirror at, let's say, point P, draw another line that joins the points F and P. We can call this line "Line X".
2. Draw another line that connects the object and F. When that line meets the concave mirror at, let's say, point Q, draw another line originating from point P and is parallel to the principal axis. We can call this line "Line Y".
3. The point where lines X and Y meet is the position where light is focused, and the image formed is inverted, magnified, and real.

You can try drawing similar ray diagrams for the following situations : When the object is beyond C, object is on C, object is on F, object is between F and O (the center of the mirror).

Correct Answer: C

Question 41

$$v = f\lambda$$

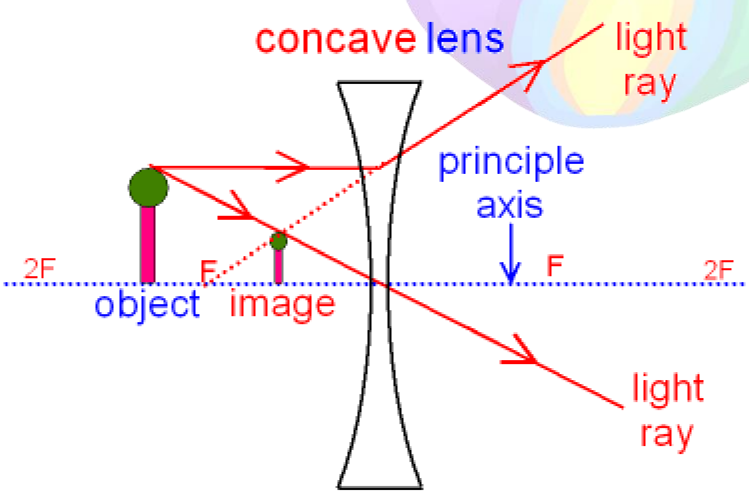
The speed v remains constant, as the speed of sound in air is constant and unique for certain physical surroundings (temperature, density and other properties of air). So, when f is doubled, λ is halved.

Note :- A lot of students assume that the speed of sound doubles, but you have to remember that the speed of sound in air is constant, only the frequency and wavelength can change.

Correct Answer: A

Question 42

A concave lens acts as a diverging lens and reflects light away from the focal point.



Just like in question 40, you can try drawing such ray diagram yourself.

Correct Answer: A

Question 43

When an object is moving in a uniform/constant velocity, it has zero acceleration. According to Newton's 2nd law, this means the object is experiencing zero net force. This means that the forces 2P and 20N acting in opposite directions must be equal in magnitude.

$2P = 20N \Rightarrow P = \frac{20}{2}N = 10N.$

When P is halved, the force acting towards the left is $\frac{P}{2} = \frac{10N}{2} = 5N$. The resultant force is 20N - 5N = 15N to the right. Applying Newton's 2nd law F=ma and substituting F = 15N and m = 5kg, we get

$a = \frac{F}{m} = \frac{15N}{5kg} = 3ms^{-2}.$

Correct Answer: B

Question 44

Pressure in a liquid is calculated by the equation

$\text{Pressure} = \rho gh$

Assume the density of liquid R is d. Theb, the density of liquid Q must be 2d.

For liquid R;

$\text{Pressure} = dgx = 4kPa$

$$5kPa = 4kPa * \frac{5}{4} = \frac{5}{4} * dgx = 1.25dgx$$

For liquid Q;

$\text{Pressure} = 1.25dgx = 2dgh$

Rearranging, we get $h = \frac{5}{8}x.$

Correct Answer: D

Question 45

$$\text{Current} = \frac{\text{Charge}}{\text{Time}}$$

In both cases, current $= \frac{2C}{20s} = 0.1A$.

Note :- In questions like this, some students tend to make mistakes such as "if the **charge density** is different, then the current should be different as well". Always keep in mind that the current depends on the **charge** and not the charge density.

Correct Answer: C

Question 46

$\text{Upthrust} = vdg$

Assuming the cross sectional area of the cube to be A, volume becomes Ah.

The volume of the cube inserted in water = A(1-0.25h) = 0.75Ah.

$\text{Upthrust} = 0.75Ah * \rho g$

The object is in equilibrium. The weight and the upthrust are the only forces acting on the object. So, according to Newton's Second Law, upthrust = weight = mg.

$\text{mass} = v\rho_s = Ah\rho_s$

$\text{Weight} = mg = Ah\rho_s g$

Since weight and upthrust are equal,

$Ah\rho_s g = 0.75Ah\rho g$

Solving, we get

$$\rho_s = 0.75\,\rho = \frac{3}{4}\rho$$

Correct Answer: D

Question 47

When two lines are parallel, their gradient is the same. The gradient of y = 5x + 3 is 5. So, the gradient of the new line must be 5 as well. The y-intercept is said to be 4. So, the new line should be (y = 5x +4) *n where n is any integer. The only answer that fits this description is A.

Correct Answer: A

Question 48

To solve the question, you have to write all the terms in powers of 3.

left hand side $= \frac{3^{2x+2}}{9} = \frac{3^{2x+2}}{3^2} = 3^{2x}$.

right hand side $= 3^{3x-1}$

Since the left and right hand sides are equal, $3^{2x} = 3^{3x-1}$.

So, $2x = 3x - 1$. Then, x = 1.

Correct Answer: C

Question 49

For the average weight of 25 people to increase by 1 kg, the total weight must increase by 25kg. The first man's weight was 60kg. So, the new man's weight must be 60kg + 25 kg = 85kg.

Correct Answer: C

Question 50

We simply have to rearrange the equations

$$T = \frac{V}{a}\sqrt{\frac{A\rho}{2F}}$$

$$\frac{Ta}{V} = \sqrt{\frac{A\rho}{2F}}$$

Squaring both sides,

$$\left(\frac{Ta}{V}\right)^2 = \frac{A\rho}{2F}$$

Making F the subject,

$$F = \frac{A\rho}{2\left(\frac{Ta}{V}\right)^2} = \frac{A\rho V^2}{2\,(Ta)^2} = \frac{A\rho}{2}\left(\frac{V}{Ta}\right)^2$$

Correct Answer: B